The Dengue Exercise

SISMID 2026 • Day 1, 11:00 • Map a search signal onto disease activity

Dengue fever is a mosquito-borne viral disease affecting millions of people in tropical regions. In this exercise you build a simple version of the digital disease-detection tool *Google Dengue Trends* for Mexico: fit a line from Google **search** interest to reported dengue **cases** on the early years, then use it to predict the later years from search alone, and judge how well a cheap, fast search signal stands in for slow official case reports.

Data. `data/MX_Dengue_trends.csv` (Mexico, monthly, 2004–2011). The `Date` column is the month. `Dengue CDC` is the number of reported dengue cases (Mexican Ministry of Health), the *ground truth*. The remaining columns (`dengue`, `sintomas de dengue`, `mosquito`, `dengue sintomas`) are Google search interest, the same kind of series you scraped in the 9:45 session.

How to work it. Two lanes lead to the same place; switch at any time.

- **Lane A (agent).** Drive a coding agent (Codex, Claude Code, or Antigravity CLI) with the prompts in `notebooks/dengue_exercise.ipynb` (they match the slides). You describe each step; the agent writes and runs the code.
- **Lane B (notebook).** Run the pre-filled `notebooks/dengue_exercise_soln.ipynb` cell by cell.

You may use **Python or R**. Materials: `https://github.com/Shihao-Yang/sismid2026`.

(a) **Plot the cases.** Plot `Dengue CDC` as a function of time. Then overlay the `dengue` search series (a second y-axis helps, since the scales differ) and print the correlation between the two columns. Do they rise and fall together? Is there a lead or lag, or any gap or spike that could fool a model?

(b) **Fit a line (least squares).** Using **only the first 36 months (2004–2006)** as training, find the best line that explains cases from the `dengue` search term:

$$y_t = a + b\,x_t + \varepsilon_t, \qquad (\hat{a}, \hat{b}) = \arg\min_{a,b} \sum_t \big(y_t - a - b\,x_t\big)^2,$$

where y_t is `Dengue CDC` and x_t is `dengue`. Report the intercept $\hat{a}$ and slope $\hat{b}$ and the in-sample R^2.

(c) **Show the fit.** Plot the fitted line on top of a scatter of cases vs. search for the training period, and compare against your plot from (a).

(d) **Predict and validate.** Using the *same* intercept and slope from (b), predict dengue cases for the **validation period 2007–2011** from the `dengue` search column **alone** (do not refit). Plot your predictions against the actual cases and report the out-of-sample correlation and RMSE. Watch for the common mistake of letting post-2006 data leak into the fit.

(e) **Discuss.** How well does the cheap search signal stand in for the slow case reports? Where does it drift? A good fit does not prove a correct model: name one reason search and cases can correlate without search *causing* anything, and one way you would guard against being fooled.

Stretch (if you finish early): dynamic training + Lasso. On the `sarahhbellum/Zika-tutorial` (`https://github.com/sarahhbellum/Zika-tutorial`) data, refit on a rolling 2-year window instead of once, and use Lasso (ℓ_1) across many search terms so the useless ones drop out:

$$\min_{a,\boldsymbol{\beta}} \sum_t \Big(y_t - a - \textstyle\sum_j \beta_j x_{j,t}\Big)^2 + \lambda \textstyle\sum_j |\beta_j|.$$